K.R. MANGALAM UNIVERSITY
THE COMPLETE WORLD OF EDUCATION

DATABASE MANAGEMENT SYSTEMS
[ENCS204]

Bachelor of Technology
in
Computer Science and Engineering

Submitted by
[LAKSHITA KALRA](2401010117)

Course Teacher:
MISS MANSI KAJAL

School of Engineering and Technology

K.R.MANGALAM UNIVERSITY
Sohna,Haryana 122103,India
JAN- MAY[2026]

AIM:

To understand and implement SQL transactions using COMMIT, SAVEPOINT and ROLLBACK. To retrieve records using SELECT queries and perform INSERT, UPDATE and ALTER operations.

DESCRIPTION:

In this experiment, I learned how SQL transactions work. I created tables and inserted records. I used COMMIT to permanently save data.
I used SAVEPOINT to create intermediate checkpoints.We used ROLLBACK to undo changes. I also practiced SELECT queries and UPDATE operations.

PROCEDURE:

STEP 1: Create CLASS Table

```
CREATE TABLE CLASS(

id INT PRIMARY KEY,

Name VARCHAR(50)

);
```

7	12:26:04	CREATE TABLE CLASS(id INT PRIMARY KEY, N...	0 row(s) affected	0.015 sec

STEP 2: Insert Initial Records

```
INSERT INTO CLASS VALUES (1,'Bravo');

INSERT INTO CLASS VALUES (2,'Alex');

INSERT INTO CLASS VALUES (4,'Cheng');
```

8	12:26:46	INSERT INTO CLASS VALUES (1,'Bravo')	1 row(s) affected	0.0054 sec
9	12:26:46	INSERT INTO CLASS VALUES (2,'Alex')	1 row(s) affected	0.00070 sec
10	12:26:46	INSERT INTO CLASS VALUES (4,'Cheng')	1 row(s) affected	0.00078 sec

STEP 3: Create EMPLOYEES Table

```
CREATE TABLE EMPLOYEES(
Employee_Id INT PRIMARY KEY,
First_Name VARCHAR(50),
Last_Name VARCHAR(50),
Email VARCHAR(50),
Phone_Number VARCHAR(20),
Hire_Date DATE,
Job_Id VARCHAR(20),
Salary INT,
Commission_Pct DECIMAL(5,2),
Manager_Id INT,
Department_Id INT
);
```

11	12:27:18	CREATE TABLE EMPLOYEES(Employee_Id INT...	0 row(s) affected	0.0098 sec

STEP 4: Create Client_master Table

```
CREATE TABLE Client_master(
ClientNO VARCHAR(10) PRIMARY KEY,
Name VARCHAR(50),
City VARCHAR(50),
```

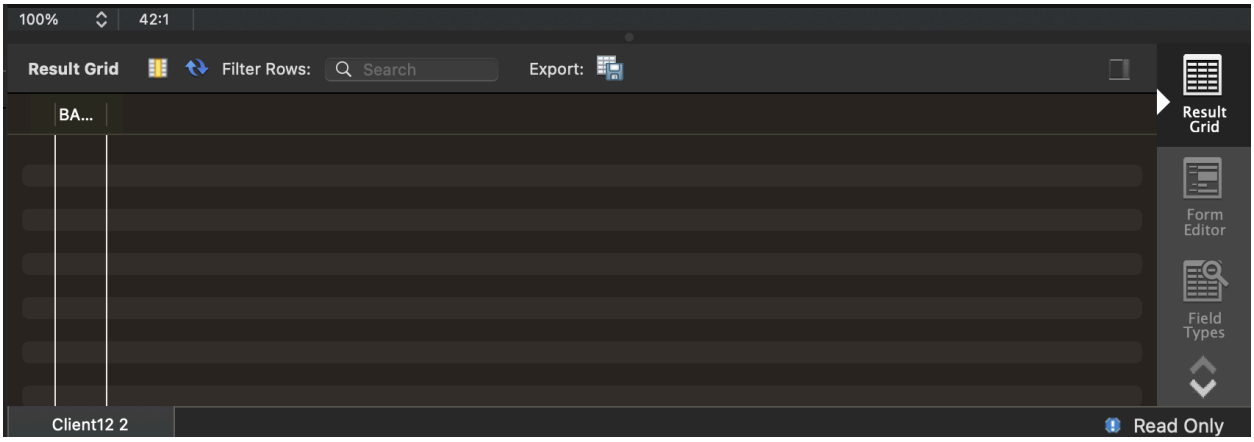
bal_due INT

);

STEP 5: Rename Table

ALTER TABLE Client_master

RENAME TO Client12;



STEP 6: Display Column Heading as BALANCE

SELECT bal_due AS BALANCE FROM Client12;

DML PART A – TRANSACTIONS (CLASS TABLE)

SQL Script Execution

INSERT INTO class VALUES (5,'Rahul');
COMMIT;

UPDATE class SET name='Abhijeet' WHERE id=5;

SAVEPOINT A;

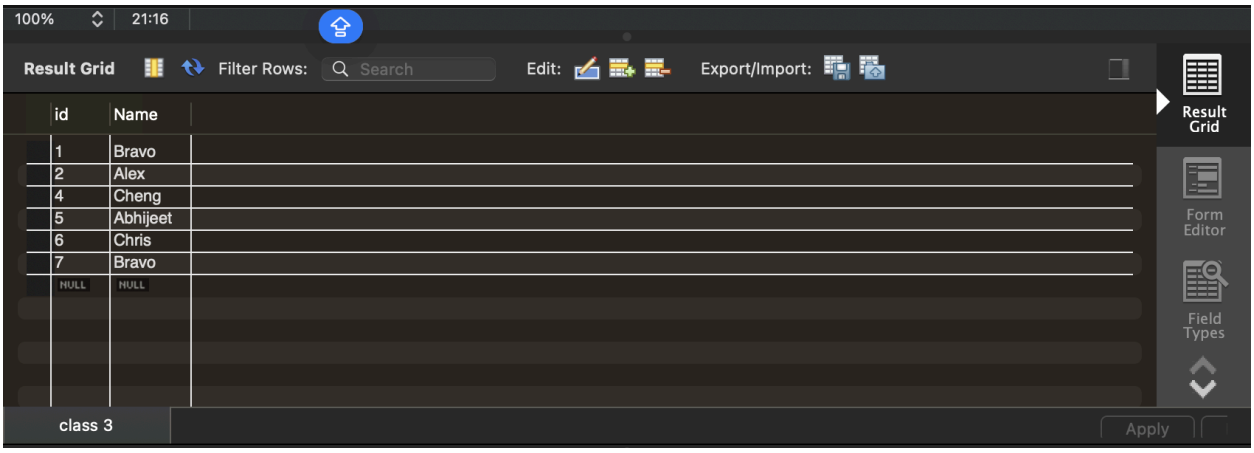
INSERT INTO class VALUES (6,'Chris');

SAVEPOINT B;

INSERT INTO class VALUES (7,'Bravo');

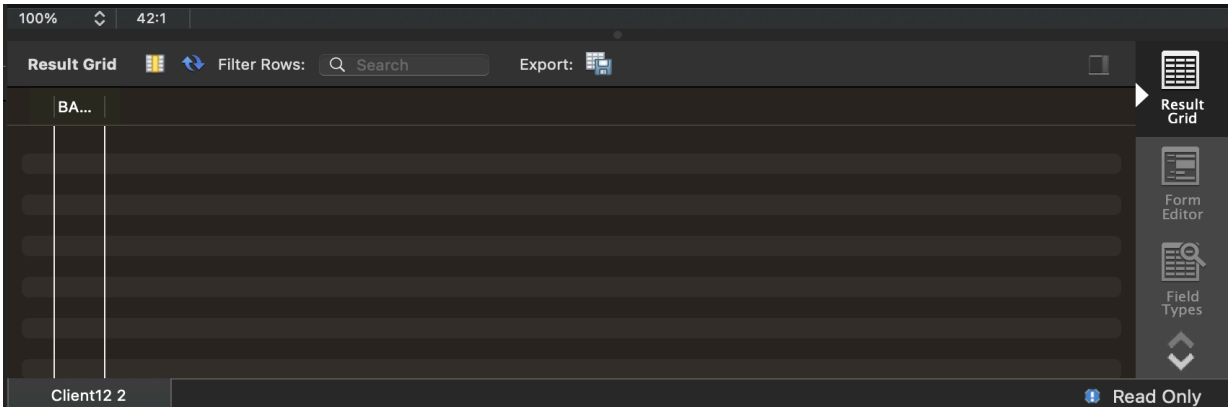
SAVEPOINT C;

SELECT * FROM class;



ROLLBACK TO B

ROLLBACK TO B;
SELECT * FROM class;



ROLLBACK TO A

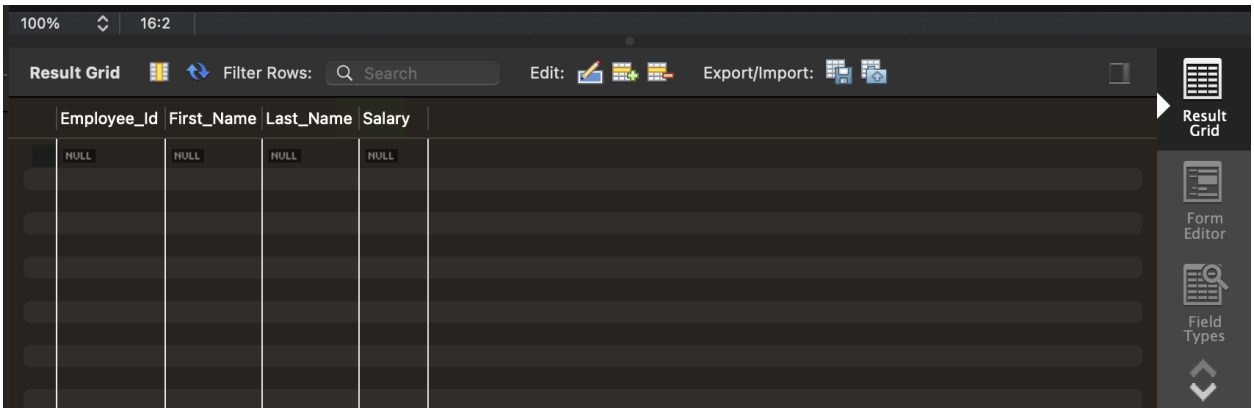
ROLLBACK TO A;

47	12:48:32	ROLLBACK TO A	0 row(s) affected	0.00032 sec

PART B – SELECT QUERIES (EMPLOYEES)

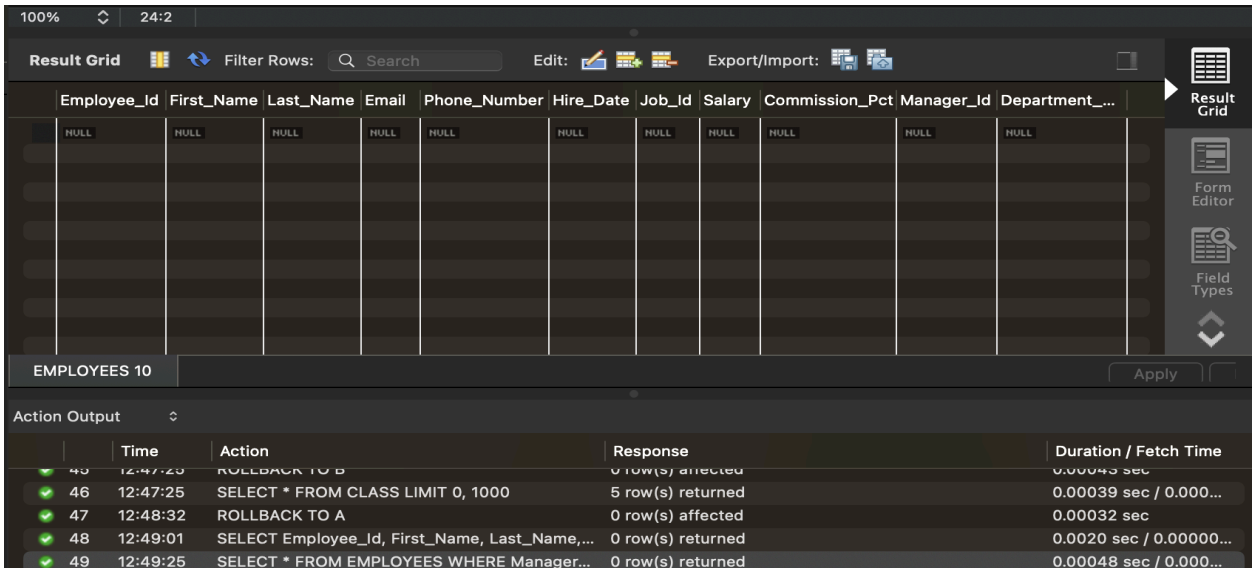
1 Employee id, names, salary

SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;



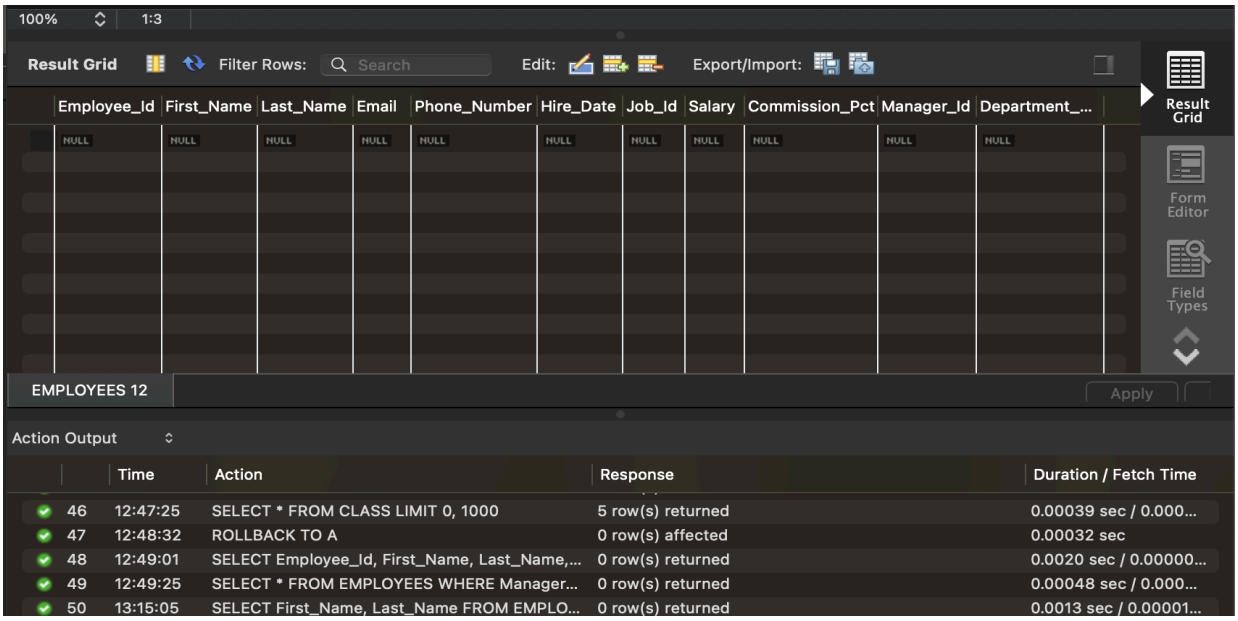
2 Employees under manager 100

SELECT * FROM EMPLOYEES
WHERE Manager_Id = 100;



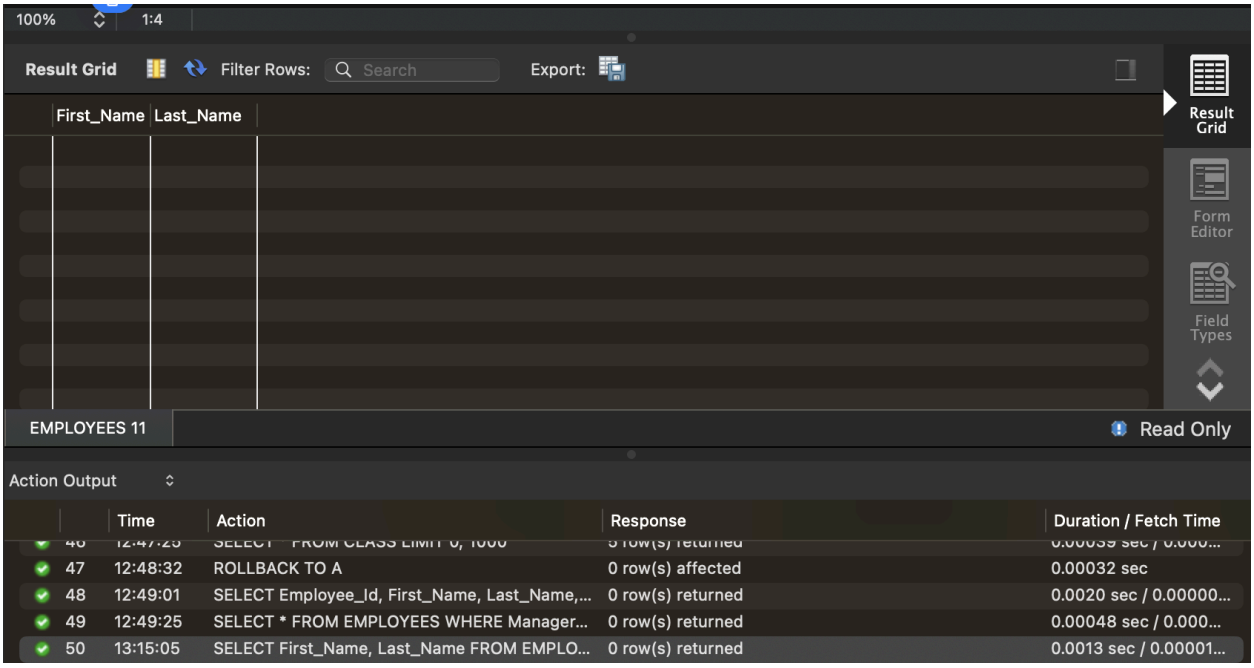
3 Salary ≥ 4800

SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Salary >= 4800;



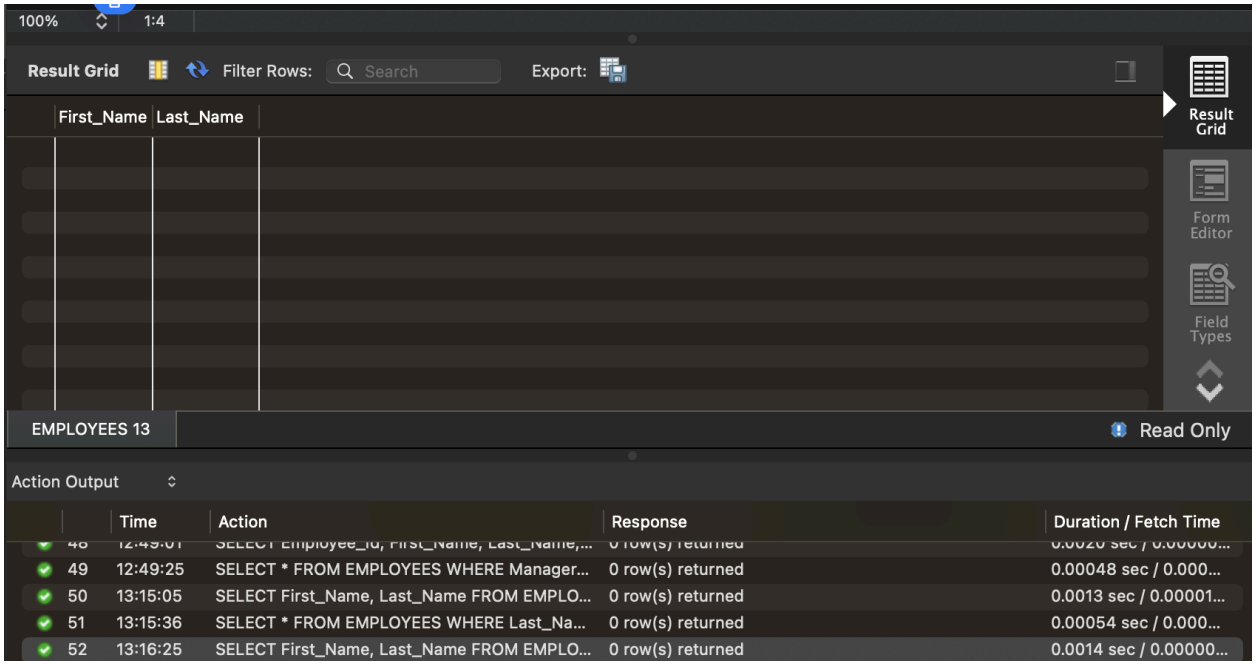
4 Last name AUSTIN

SELECT * FROM EMPLOYEES
WHERE Last_Name='AUSTIN';



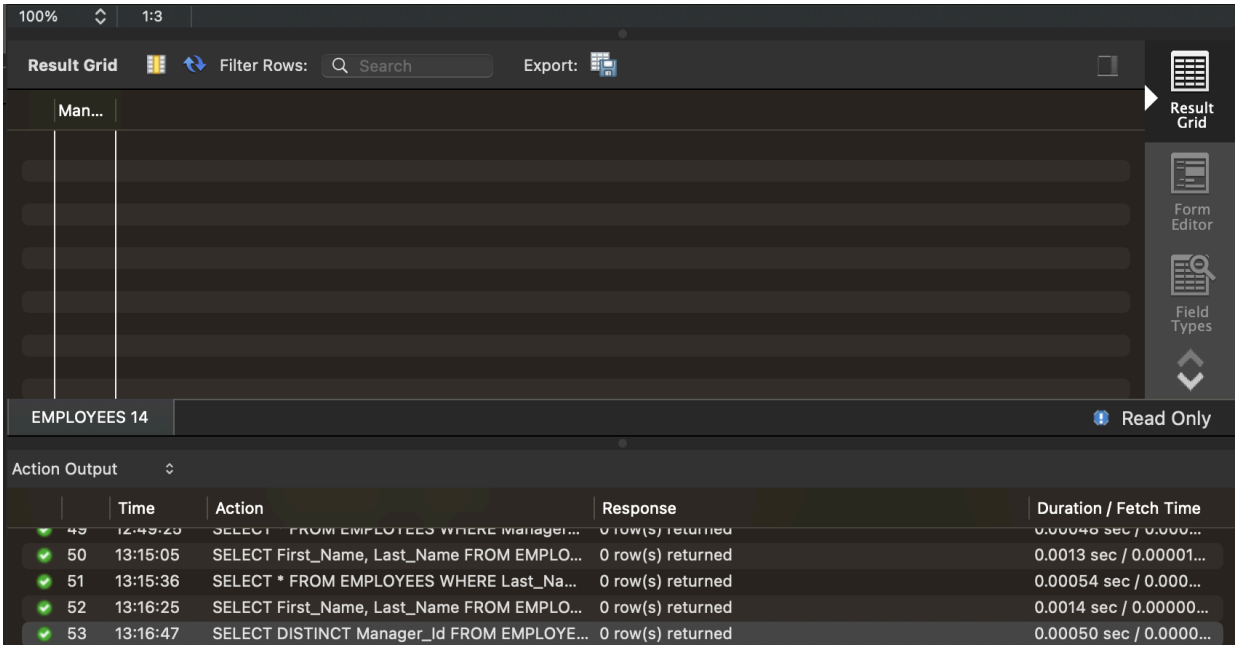
5 Departments 60,70,80

SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Department_Id IN (60,70,80);



6 Unique Manager_Id

SELECT DISTINCT Manager_Id
FROM EMPLOYEES;



PART C – Client_master Table

1 Insert 5 Records

INSERT INTO Client12 VALUES ('C123','Rahul','Delhi',6000);
INSERT INTO Client12 VALUES ('C124','Amit','Mumbai',3000);
INSERT INTO Client12 VALUES ('C125','Neha','Noida',7000);
INSERT INTO Client12 VALUES ('C126','Simran','Pune',4500);
INSERT INTO Client12 VALUES ('C127','Karan','Jaipur',8000);

55	13:17:17	INSERT INTO Client12 VALUES ('C124','Amit','M...	1 row(s) affected	0.00022 sec
56	13:17:17	INSERT INTO Client12 VALUES ('C125','Neha','N...	1 row(s) affected	0.00017 sec
57	13:17:17	INSERT INTO Client12 VALUES ('C126','Simran','...	1 row(s) affected	0.00013 sec
58	13:17:17	INSERT INTO Client12 VALUES ('C127','Karan','J...	1 row(s) affected	0.00022 sec

2 bal_due > 5000

SELECT Name FROM Client12
WHERE bal_due > 5000;

Name
Rahul
Neha
Karan

Result Grid
Form Editor
Field Types
Client12 15
Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
56	13:17:17	INSERT INTO Client12 VALUES ('C125','Neha','N...	1 row(s) affected	0.00017 sec
57	13:17:17	INSERT INTO Client12 VALUES ('C126','Simran','...	1 row(s) affected	0.00013 sec
58	13:17:17	INSERT INTO Client12 VALUES ('C127','Karan','J...	1 row(s) affected	0.00022 sec
59	13:17:45	SELECT Name FROM Client12 WHERE bal_due...	3 row(s) returned	0.00047 sec / 0.0000...

3 Update bal_due of C123

UPDATE Client12

```
SET bal_due = 5100
WHERE ClientNO='C123';
```

60	13:18:14	UPDATE Client12 SET bal_due = 5100 WHERE...	1 row(s) affected Rows matched: 1 Changed: 1 Warni...	0.0023 sec

CONCLUSION:

In this experiment, I understood how SQL transactions work.I learned how COMMIT permanently saves data.SAVEPOINT helps in creating checkpoints. ROLLBACK helps in undoing changes.I also practiced retrieving and updating records in SQL.